SONY

Color Digital Camera

DXC-S500



Product Information Manual

Internal Use Only

1. INTRODUCTION

As the number of microscopy applications increases, the method of capturing images has changed from using NTSC/PAL-based video signals to the simplicity and convenience of PC-based image data. A flexible interface with PCs along with superior picture quality are now essential to microscopy, and the Sony DXC-S500 Digital Camera System meets these demands by providing both.

Comprising two separate units, the DXC-S500 includes a compact and lightweight camera head unit (CHU) and a camera control unit (CCU) with an integrated 3.5 type LCD monitor.

The 2/3 type five mega-pixel CCD incorporated in the CHU provides ultra-high-resolution images and precise color reproduction. The LCD on the CCU enables you to monitor live images and offers user-friendly operation.

The added advantage of the DXC-S500 is its flexible interface with PCs. An IEEE 1394 interface on the CCU allows uncompressed image data transfer to a PC. A built-in PCMCIA card slot enables you to store images on various media, such as "Memory Stick™", CompactFlash™, SmartMedia™ and removable hard disk drives.

What's more, a hot-plug function allows the camera to be effortlessly connected to a PC, while the supplied TWAIN driver supports various application software such as Photoshop® and Image-Pro® Plus.

With excellent picture quality, sophisticated features and PC-friendly operation, the Sony DXC-S500 is the ideal system to support your medical and scientific microscopy applications.



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2. PRODUCT CONCEPT

Ultra-high-resolution Color Digital Camera for Microscopy



- **Superb picture quality with a 2/3 type five mega-pixel CCD**
- **Flexible interface with PCs**
  - **IEEE 1394 interface**
  - **PC card slot**
  - **TWAIN driver**
- **User-friendly operation with an integrated 3.5 type LCD monitor**
- **C-mount system**

3. APPLICATIONS

Bio-medical Microscopy

- **Microbiology**
- **Pathology**
- **Genetics**

4. KEY FEATURES

Superb Picture Quality

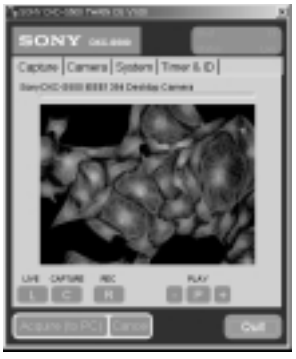
By utilizing several core technologies, the DXC-S500 offers superb picture quality. With a five mega-pixel 2/3 type IT CCD, the DXC-S500 can capture ultra-high-resolution and high-precision images. The maximum number of pixels used for image capture is 2560 (H) x 1920 (V), providing superior resolution. The use of RGB primary color mosaic filters contributes to precise color reproduction, while a cooling device cools the CCD chip to 5 ˚C (at 0 to 35 ˚C ambient temperature) to achieve excellent signal-to-noise ratio.

Long-term Exposure Shooting

Built-in frame memory provides a remarkable enhancement in sensitivity by allowing the charge accumulation period of the CCD (typically 1/60 second) to be extended to a maximum of 60 seconds.

Fluorescence Image Capturing

The DXC-S500 provides a fluorescent mode (FL MODE), which facilitates focus and capture adjustments for dark objects at the touch of a button. When activated, it automatically boosts the camera gain level and lowers the frame rate so the dark object can be clearly seen. When focus adjustments have been made, pressing the "CAPTURE" button resets the gain and calculates the appropriate shutter speed (Long-term Exposure) so a clear image can be obtained. And, thanks to the CCD cooling device, the dark noise that is typically seen in Long-term Exposure mode is dramatically reduced.



IEEE 1394 Interface

The IEEE 1394 (400 Mb/s) interface located on the camera control unit enables direct interface with various PCs or IEEE 1394 boards*. The interface supports a hot-plug function for quick access to the camera's image without restarting the PC. The pictures are transmitted in uncompressed IEEE 1394 format for highest accuracy.

* When Microsoft® Windows® is used, IEEE 1394 boards conforming to OHCI are supported. For Macintosh®, genuine FireWire® boards are supported.



PC Card Slot

A PC card slot is located on the front of the CCU for storing images directly onto various media. This slot supports PCMCIA ATA Type II*, enabling data storage in a "Memory Stick", CompactFlash, SmartMedia and removable hard disk drive.

* PC cards supporting 3.3 V power supply only are not available.



TWAIN Driver

The supplied TWAIN driver enables the use of various application software. You can use this driver to monitor live images on the PC application window, and to capture images with a simple mouse click.



Selectable Image Size and Recording Format

Three recording image sizes are available for selection, and images of each size can be stored in TIFF (uncompressed) or JPEG (compressed, three modes) format.

Image recording size (H x V)	TIFF (uncompressed)	JPEG (compressed)		
		High	Mid	Low
FULL: 2560 x 1920	15 MB	5 MB	2.5 MB	1.3 MB
SXGA: 1280 x 960	3.8 MB	1.3 MB	630 KB	310 KB
VGA: 640 x 480	940 KB	310 KB	160 KB	78 KB

The above image data sizes are approximations.

Analog Video Output

Analog composite output (BNC) and Y/C output (S-Video) terminals allow high-frame-rate live monitoring. The output signal supports VGA (640 x 480) size at a 30 fps (NTSC)/25 fps (PAL) frame rate, so you can easily focus the microscope by viewing the image on a video monitor.

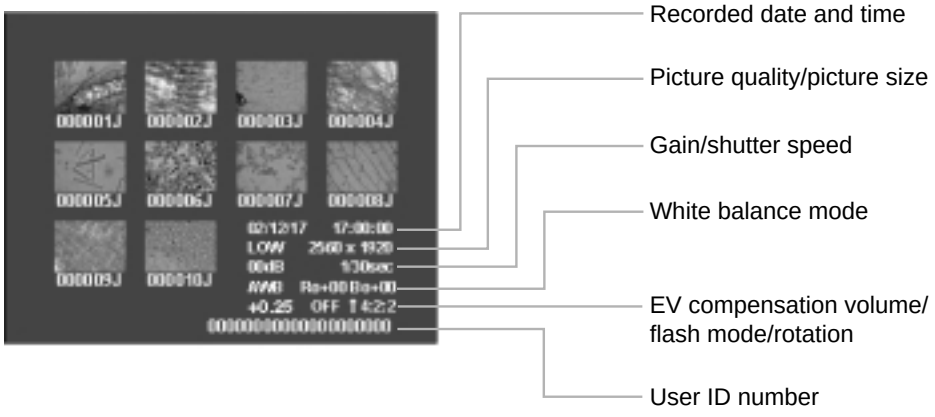
User-Friendly Operation

A 3.5 type 200,000-pixels LPS TFT LCD monitor on the CCU provides sophisticated operability, including image capture, index picture display and menu setting. You can also monitor the live image with the built-in LCD monitor.



Index Picture Display

The Index Picture Display allows images stored on the PC card to be previewed as thumbnails. Up to 10 thumbnail images can be viewed at any one time. This allows the user to easily check, playback, and erase stored images.



AE Functions

The AE (Auto Exposure) function automatically detects the lighting conditions or brightness of the object and adjusts the shutter speed and gain for proper exposure. The DXC-S500 is equipped with a number of AE modes that offer superb image quality.

C-Mount System

To provide an extensive choice of lenses, the DXC-S500 uses the 2/3 type C-mount system for flexible and easy lens mounting.

Foot Switch

An optional FS-20 foot switch is available, enabling you to capture and record images by simply stepping on the switch.

Histogram Display

The DXC-S500 automatically detects the luminance distribution of images, displaying a histogram on the monitor so you can verify the condition of each exposure.

User File (max. three files)

A variety of customized data can be saved as a user file, up to three of which can be stored in the CCU. By loading this user file, you can quickly recreate particular setup conditions.

Black & White Mode

The image output can be selected from either Color or Black & White modes.

Interval Recording (max. 24 hours)

It is possible to record at pre-determined intervals (up to 24 hours).

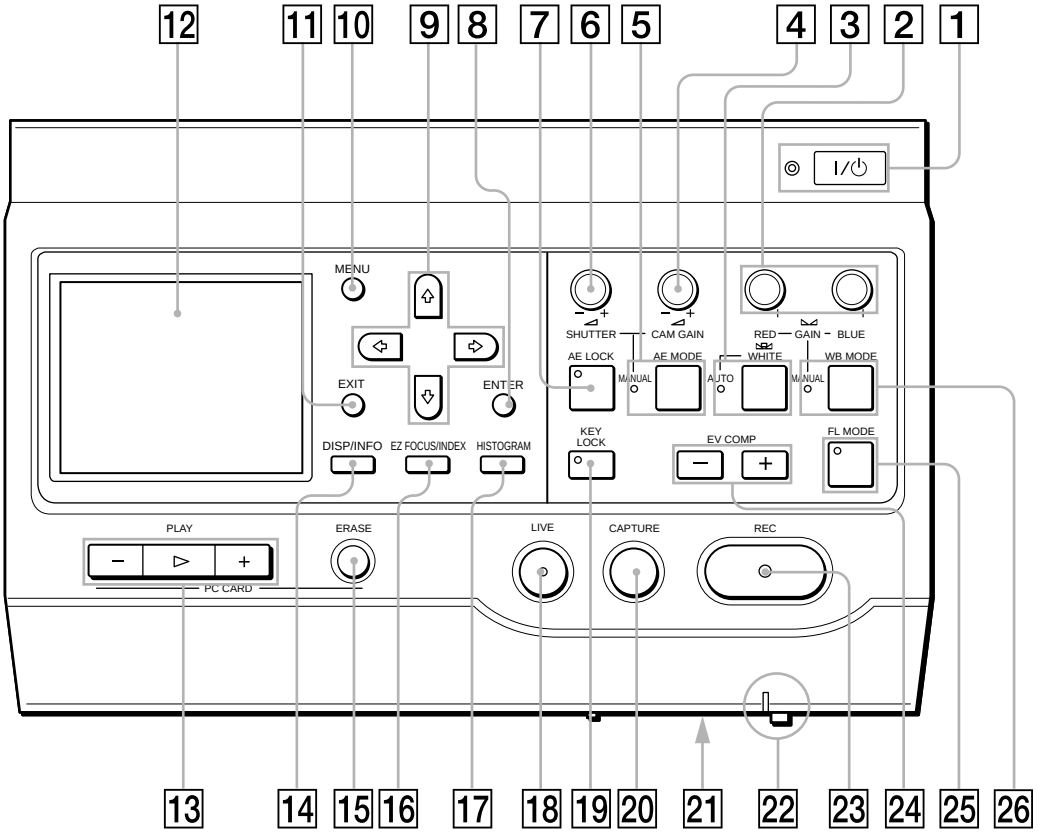
Picture Rotation (0˚/180˚ selectable)

This function rotates the image by 0˚ to 180˚.

2x Digital Zoom

5. PANEL LAYOUT

Camera Control Unit (CCU) Control Panel



1 On/standby switch and indicator

When the power of the unit is on, the indicator is lit in green, and when the unit is in standby mode the indicator is lit in orange.

2 GAIN (BLUE/RED) controls

Adjusts the red and blue gain levels in manual white balance mode, or after executing the Auto White Balance mode (WB-AUTO).

3 WHITE (white balance) button and AUTO indicator

When the WB MODE is set to AUTO (the AUTO indicator is lit in green), pressing the WHITE button activates WB-AUTO (Auto White Balance).

4 CAM GAIN (camera gain) control

In MANUAL exposure mode (the MANUAL indicator is lit in red), turning the CAM GAIN volume adjusts the camera gain level within the range of 0 dB to 12 dB.

5 AE MODE button and MANUAL indicator

Sets the exposure mode of the camera to AE-AUTO or MANUAL. In AE-AUTO mode, the electronic shutter speed and the camera gain automatically change in accordance with the brightness of the subject. In MANUAL mode (the MANUAL indicator is lit in red), the electronic shutter speed and the camera gain can be adjusted independently.

6 SHUTTER (electronic shutter) control

In MANUAL exposure mode (the MANUAL indicator is lit in red), turning the SHUTTER volume controls the shutter speed within the range of 1/10,000 s to 60 s.

7 AE LOCK button and indicator

In AE-AUTO mode, pressing the AE LOCK button fixes the exposure time (the AE LOCK indicator is lit in green).

8 ENTER button

In MENU DISPLAY mode: Displays the next layer of the menu system.
In INDEX DISPLAY mode: Displays the selected image in full screen.
In Camera LIVE mode: Initializes the focus indicator display.

9 ↑/↓/←/→ buttons

In MENU DISPLAY mode: The ↑ (cursor up) or ↓ (cursor down) button moves the cursor in the menu display up or down. The ← (data down) or → (data up) button sets the data in the menu display.
In INDEX DISPLAY mode: Selects an image from the thumbnail images displayed.

10 MENU button

In Camera LIVE mode: Displays the MAIN MENU on the LCD monitor.
In PLAY mode: Displays the menu for replay.

11 EXIT button

When the menu is displayed, pressing the EXIT button restores the previous menu.

12 LCD monitor

Displays the picture being shot, a captured image, or a playback image from a PC card. Also displays the menu, the list of still images (INDEX DISPLAY), or information such as shutter speed, file name, and picture size.

13 PLAY ▸/+/– buttons

▸ **button:** Replays a still image recorded on a card.
+ button: Replays the still image of the next file number.
– **button:** Replays the still image of the previous file number.

14 DISP/INFO (display/information) button

In Camera LIVE mode: Displays information on the camera's live image such as white balance mode, EV compensation volume, and shutter speed.
In PLAY mode: Displays information about the image recorded to a PC card, such as record date and time, EV compensation volume, and shutter speed.

15 ERASE button

In PLAY mode: Erases the files on the loaded PC card.
In INDEX DISPLAY mode: Erases the selected image from thumbnail images.

16 EZ FOCUS (easy focus)/INDEX button

In Camera LIVE mode: Activates 2x electronic zoom to facilitate focusing operations.
In PLAY mode: Displays thumbnail images of still pictures (INDEX DISPLAY) recorded to a PC card.

17 HISTOGRAM button

In Camera LIVE mode: Displays a histogram of the image on the LCD monitor.

18 LIVE button and indicator

Activates the Camera LIVE mode (the indicator is lit in green).

19 KEY LOCK button and indicator

Pressing the KEY LOCK button for more than two seconds (the indicator is lit in green) disables the operation of buttons and volumes.

* The KEY LOCK mode retains while the unit is switched off.

20 CAPTURE button

In Camera LIVE mode: Pressing the CAPTURE button stores a picture in the built-in frame memory of the camera.

21 Card slot

Accepts a PC card or a Memory Stick installed in the MSAC-PC2 Memory Stick adaptor.

22 CARD EJECT button and access lamp

Pressing this button ejects the loaded PC card. The access lamp is lit in red during reading or writing.

* Do not eject the card while the access lamp is lit.

23 REC (record) button

Records a still image to a PC card. The access lamp is lit in red during recording.

24 EV COMP (compensation) +/– buttons

In AE-AUTO or FL MODE, pressing the + or – button adjusts the exposure compensation.

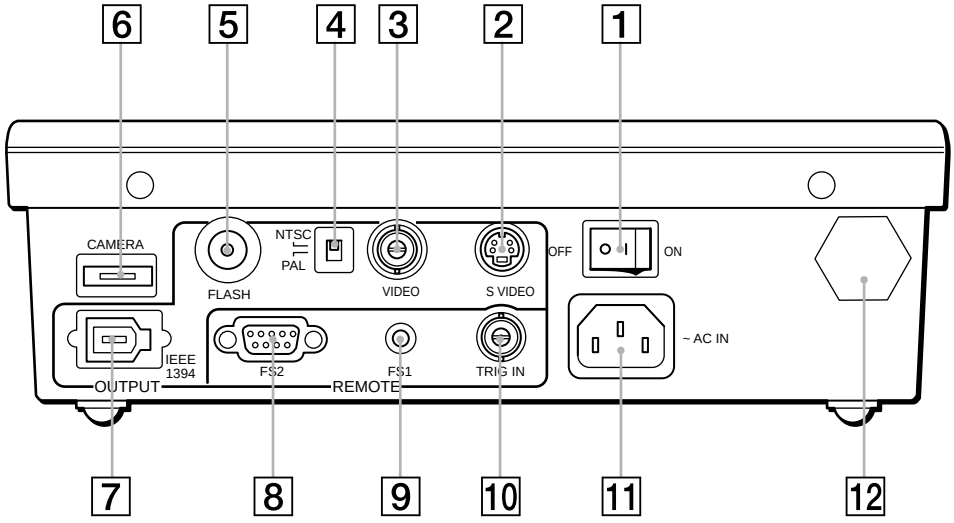
25 FL (fluorescent) MODE button and indicator

Activates the fluorescent mode of the camera. When activated, the indicator is lit in green.

26 WB (white balance) MODE button and MANUAL indicator

Sets the White Balance adjustment mode to AUTO or MANUAL. In WB-AUTO mode, pressing the WHITE button automatically adjusts the white balance. In WB-MANUAL mode (the MANUAL indicator is lit in red), the white balance is adjusted manually with the GAIN RED/BLUE volumes.

Camera Control Unit (CCU) Rear Panel



1 O/I (main power) switch

Switches the main power of the camera on or off.

2 S-VIDEO OUTPUT connector (mini DIN 4-pin)

Outputs an S-VIDEO (Y/C video) signal.

3 VIDEO OUTPUT connector (BNC-type)

Outputs a composite video signal.

4 NTSC/PAL select switch

Selects the output video signal from NTSC or PAL.

5 FLASH connector

Connects the cable of a flash lamp.

6 CAMERA connector (10-pin)

Connects to the CAMERA connector on the CHU using the supplied camera cable.

7 IEEE 1394 (digital interface) connector * (6-pin)

Connects to a computer via an IEEE 1394 cable (not supplied).

* This connector cannot be used for connection to camera equipment with a DV connector.
* This connector is not compatible with bus power.

8 FS2 connector (D-sub 9-pin)

Reserved for future use.

9 FS1 connector (stereo mini jack)

Connects to the optional FS-20 foot switch.

10 TRIG IN (trigger input) connector (BNC-type)

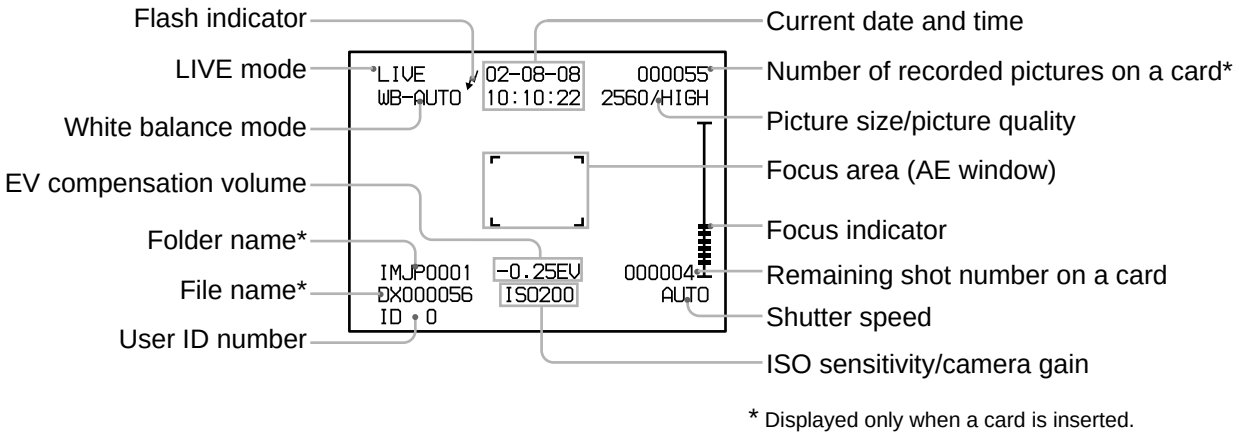
Used when an external device, such as a slave unit, generates the trigger signal. When using a flash lamp in INPUT mode, connects a slave unit to the TRIG IN connector.

11 ~AC IN socket

Connects to the AC power cord.

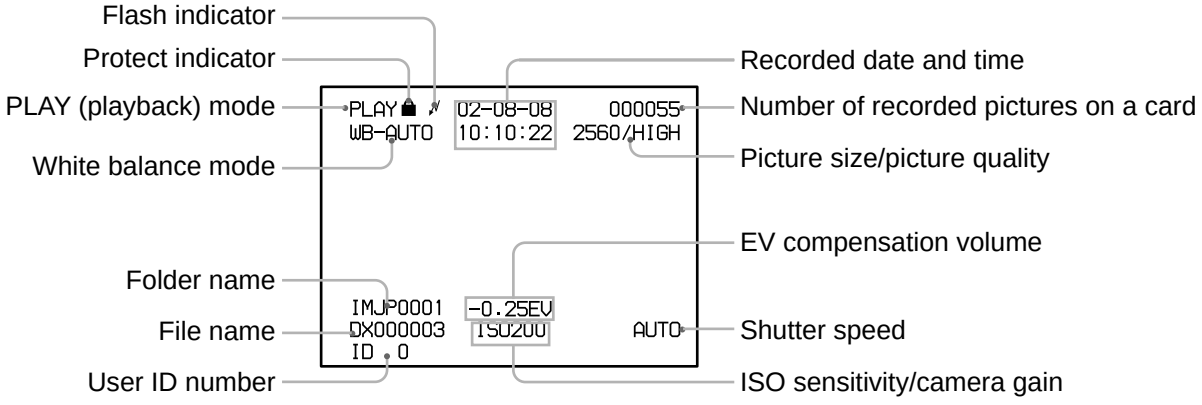
12 Ventilation opening

LCD Monitor Display Camera LIVE Mode



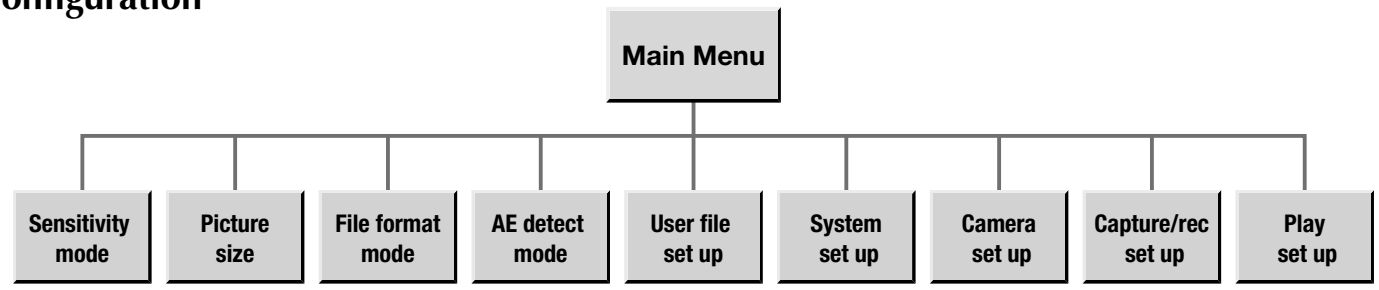
* Displayed only when a card is inserted.

LCD Monitor Display PLAY Mode

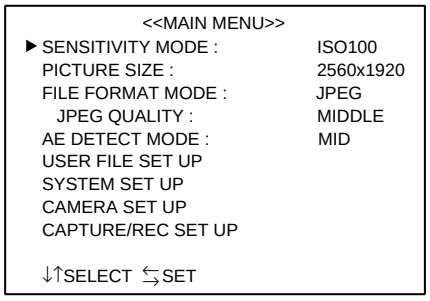


6. MENU FUNCTION

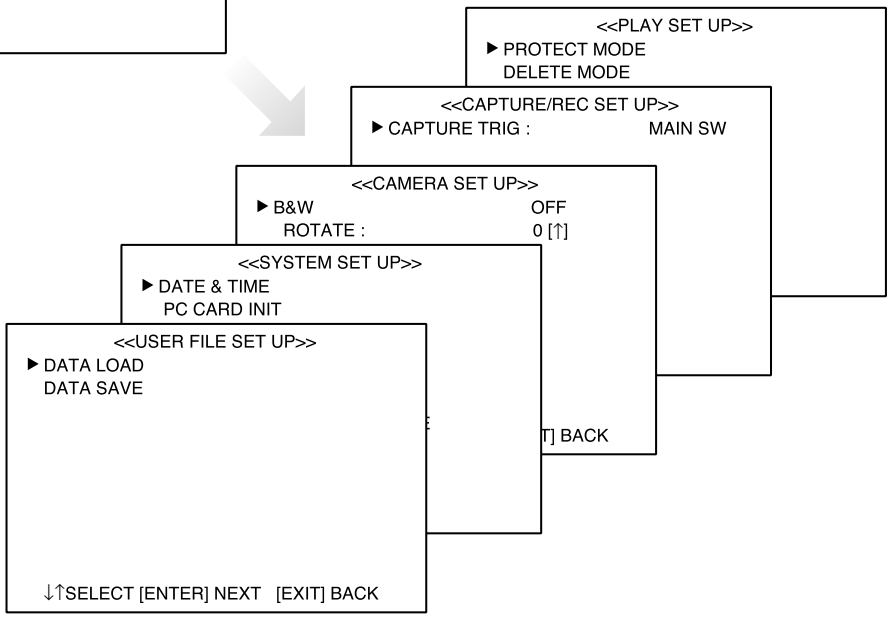
Configuration



Main Menu

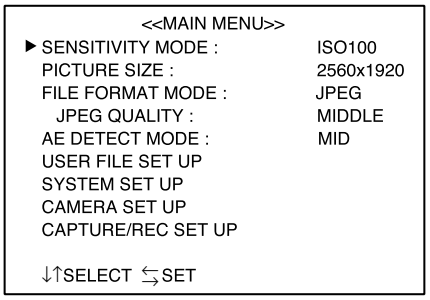


Sub Menu



Setting Items in the Menu

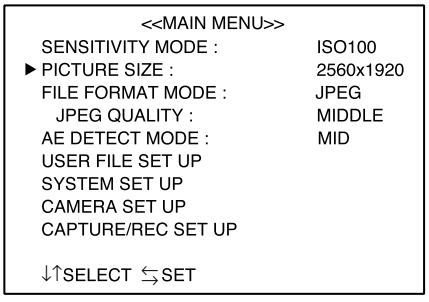
Recording Mode Menus



SENSITIVITY MODE:

Sets the gain used for image capture.

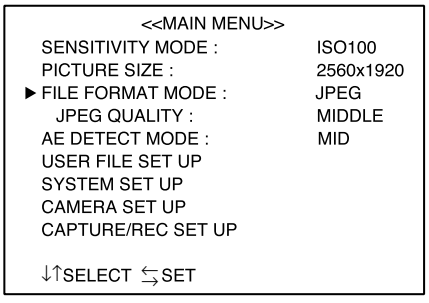
- In AE-AUTO mode, the sensitivity is selectable from ISO100, ISO200, and ISO400.
- In MANUAL exposure mode, the gain value adjusted with the CAM GAIN volume on the CCU is displayed.
- In FL MODE, the fluorescent mode is activated.



PICTURE SIZE:

Selects the size of the image to be captured.

- 640 x 480
- 1280 x 960
- 2560 x 1920

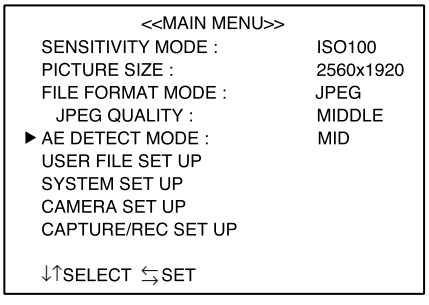


FILE FORMAT MODE:

Selects the file format of an image to be recorded to a PC card.

- JPEG (compressed)
- TIFF (uncompressed)

  When JPEG is selected as the FILE FORMAT, the data compression ratio is selectable from three modes – LOW, MIDDLE, or HIGH.

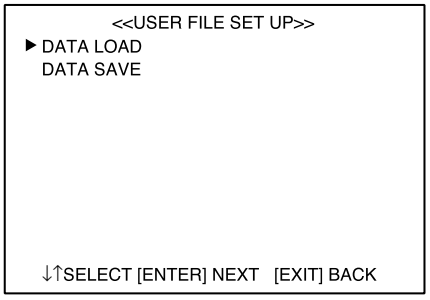


AE DETECT MODE:

Sets the size of the AE window.

- SPOT
- MID
- LARGE

USER FILE SET UP Menu



DATA LOAD:
Selects the user setting to be loaded.

- No.0 (RESET): the factory settings
- No.1 (yy.mm.dd): user file 1
- No.2 (yy.mm.dd): user file 2
- No.3 (yy.mm.dd): user file 3

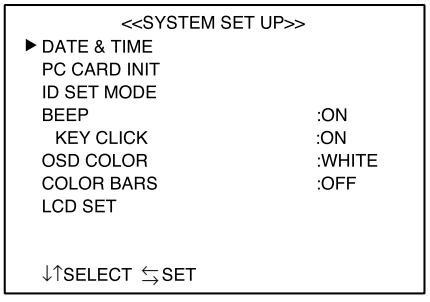
  The year, month, and day that the data was saved are indicated in parentheses.

DATA SAVE:
Saves the user setting as a user file.

- No.1 (yy.mm.dd): user file 1
- No.2 (yy.mm.dd): user file 2
- No.3 (yy.mm.dd): user file 3

  The year, month, and day that the data is to be saved are indicated in parentheses.

SYSTEM SET UP Menu



DATE & TIME:
Sets the date and time.

PC CARD INIT:
Initializes the PC card to be used with the camera.

ID SET MODE:
Assigns a user ID number up to 20 digits long.

BEEP:
Selects whether a beep is sounded during operation (ON/OFF selectable).

- KEY CLICK: Selects whether to sound a beep when an operation button is pressed (ON/OFF selectable). This item is displayed only when BEEP is set to ON.

OSD COLOR:
Selects the color of the characters used for menus, information, and messages displayed on the LCD (WHITE/BLUE/RED selectable).

COLOR BARS:
Outputs a color bar signal to the LCD monitor or an external monitor (ON/OFF selectable).

LCD SET:
Adjusts the LCD monitor of the CCU.

- CONTRAST/BRIGHT/COLOR/BACKLIGHT adjustments are available.

Setting Items in the Menu

CAMERA SET UP Menu

```
<<CAMERA SET UP>>
▶ B&W                    OFF
  ROTATE :               0 [⇑]
  SHARPNESS LEVEL :      ±00
  AUTO-WHITE-PAINT       OFF
  GAMMA :                ON
  AE SPEED :             NORMAL
  FLASH MODE :           OFF
  DISPLAY SET :

  ↓↑SELECT [ENTER] NEXT  [EXIT] BACK
```

B&W*: Displays the Camera LIVE picture in black and white.

- ON: Black and white

  *When B&W is selected, the image will also be captured in B&W.

- OFF: Color

ROTATE: Rotates the image by 180˚ (0˚/180˚selectable).

SHARPNESS LEVEL:
Adjusts the sharpness of an image. Pressing the ← button softens the sharpness of an image, while pressing the → button sharpens it.

AUTO-WHITE-PAINT:
Allows an offset to be added to the auto white balance convergent point using the GAIN RED/BLUE volumes on the CCU. This offset can be turned ON or OFF.

GAMMA: Selects the gamma compensation.

- OFF: Gamma compensation off.
- ON (MS): Gamma compensation for microscope shooting.
- ON (STD): Gamma compensation for a CRT monitor.

AE SPEED: Selects the focusing speed in AE mode.

- NORMAL
- FAST
- SLOW

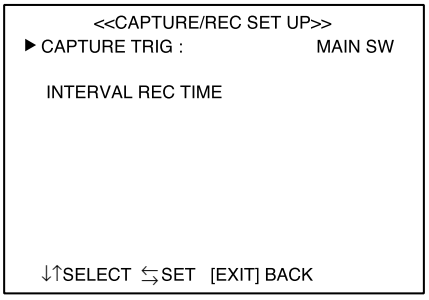
FLASH MODE: Selects the shooting mode when using a flash.

- OFF: Deactivates the flash mode.
- EXT: INPUT: Selected when using a slave unit. The flash lamp is synchronized with the trigger signal from a slave unit.
- EXT: OUTPUT: Selected when using a flash lamp. The flash lamp is synchronized with the trigger signal from the CCU.

DISPLAY SET (in Camera LIVE mode):
Selects the items that are displayed on the monitor when pressing the DISP/INFO button on the CCU.

- SIZE/FORMAT/QLTY (Picture size/recording format/picture quality).
- FOLDER/FILE NAME (Folder name/file name of an image recorded to a PC card).
- ID NO (User ID number).
- REMAINING SHOT (Remaining number of shots that can be recorded to a PC card).
- TV/GAIN/EV COMP (Shutter speed/gain/EV compensation volume).
- AE FRAME (AE window).
- FOCUS DISPLAY (Focus indicator).
- WB WHITE (White Balance mode).
- DATE & TIME (Current date and time).

CAPTURE/REC SET UP Menu



CAPTURE TRIG:

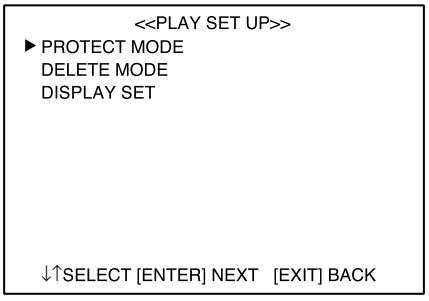
Selects the input connector to be used for image capture control.

- MAIN SW: Selected when capturing images with the CAPTURE button on the CCU.
- FS1: Selected when capturing images by switching the FS-20 foot switch connected to the FS1 connector.
- FS2: Selected to activate the input signal from the FS2 connector.
- EXT TRIG: Selected when using a slave unit connected to the TRIG IN connector.

INTERVAL REC TIME:

Sets an interval time for recording. The interval time is selectable within the range of 10 seconds to 24 hours.

PLAY SET UP Menu



PROTECT MODE:

Protects the images recorded on a PC card from accidentally being erased.

- 1 SHOT ON: Protects the image being played.
- 1 SHOT OFF: Removes protection from the image being played.
- ALL ON: Protects all images recorded on the PC card.
- ALL OFF: Removes protection from all images recorded on the PC card.

DELETE MODE:

Deletes images recorded on the PC card.

- 1 SHOT DELETE: Deletes the image being played.
- ALL DELETE: Deletes all images recorded on the PC card.

DISPLAY SET (in CAPTURE/PLAY mode):

Selects items to be displayed on the monitor when pressing the DISP/INFO button on the CCU.

- SIZE/FORMAT/QLTY (Picture size/recording format/picture quality).
- FOLDER/FILE NAME (Folder name/file name of an image recorded on a card).
- ID NO (User ID number).
- REMAINING SHOT (Remaining number of shots that can be recorded on a PC card).
- DATE & TIME (Date and time of recording).
- TV/GAIN/EV COMP (Shutter speed/gain/EV compensation volume).
- WB MODE (White Balance mode).

7. TWAIN DRIVER SOFTWARE

By opening the supplied TWAIN driver software from your application software (e.g., Photoshop or Image-Pro Plus), images stored on the camera can be transmitted to a computer in uncompressed IEEE 1394 format. You can also operate basic functions and alter settings of the camera from the computer.

* The supplied TWAIN driver software can only be used with the Sony DXC-S500 color digital camera.



System Requirements for the Supplied TWAIN Driver Software

For Microsoft Windows users

Hardware

IBM PC/AT®-compatible machine

CPU:	Intel® Pentium® III or faster (Pentium III 400 MHz or faster recommended)
Memory capacity:	128 MB RAM minimum (more than 256 MB RAM recommended)
Cache memory:	256 KB minimum
Hard disk:	50 MB memory or greater required
PCI slot:	One free PCI slot required (PCI 2.1-compliant)
Video board:	24-bit full-color video board supporting DirectX V8.0 or later required. To display images correctly with this software, the video board requires the ability to show at least 16,000,000 colors. To display live images correctly, the video board requires a high-performance video chip supporting DirectX, such as ATI RADEON 7500/8500 series or NVIDIA GeForce 2/3/4 series.

*The TWAIN driver software displays the live image using the function of the high-performance video board through the DirectX V8.0. Correct operation of the software is not guaranteed with some notebook computers in which the video board cannot be changed.

Software

Supporting OS:	Windows 98SE/ME/2000 Professional SP2/XP Home/XP Professional
DirectX:	DirectX Runtime module V8.0 or later
Application software supporting TWAIN driver:	Photoshop V5.0 or later Image-Pro Plus V4.5 or later

For Macintosh users

Hardware

Apple Computer, Inc. genuine personal computer

CPU:	PowerPC G3/G4 PowerMac, PowerBook, iMac, iBook series
Memory capacity:	128 MB RAM minimum (more than 256 MB RAM recommended)
Hard disk:	50 MB memory or greater required
FireWire port:	One free built-in FireWire port required

Software

Supporting OS:	Mac OS 8.6, 9.0, 9.1, 9.2
QuickTime:	QuickTime V4.0 or later
Application software supporting TWAIN driver:	Photoshop V5.0 or later

Locations and Functions of Controls (for Windows)

Image Information Panel



The Image Information Panel displays information about the images.

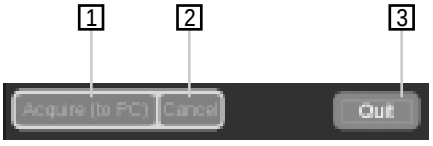
1 Shot number display

Displays the number of shots taken by the camera.

2 Status display

Displays the status of the DXC-S500 camera.

Image Control Panel



The Image Control Panel contains buttons for transmitting the image from the camera.

1 Acquire (to PC) button

Clicking this button transmits the images from the CCU into the application software. The transmitted image then appears in the application software window.

2 Cancel button

Clicking this button cancels transmission of the image into the application software.

3 Quit button

Clicking this button exits the TWAIN driver software and returns to the application software.

Locations and Functions of Controls (for Windows)

Capture Tab



The Capture tab is used to capture or record images.

1 Device display area

Shows the device name recognized.

2 Progress bar display area

Shows the progress bar when an image is being transferring.

3 Image display area

Shows a live image, the index display or a playback image.

4 LIVE L button

Selects the Camera LIVE mode to display the live image on the LCD or video monitor.

5 CAPTURE C button

Captures a still image in the built-in frame memory of the camera.

6 REC R (record) button

Records a still image to the PC card inserted in the CCU.

7 − button

Selects a still image of the previous file number on the PC card.

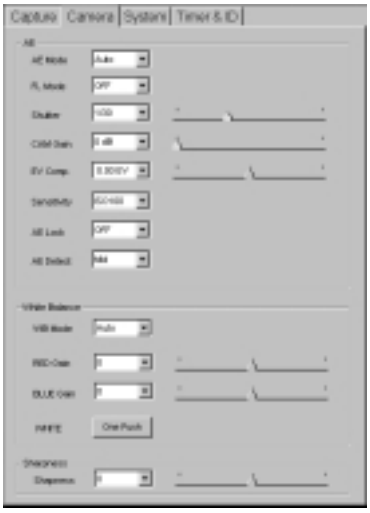
8 PLAY P button

Plays a still image stored on the PC card.

9 + button

Selects a still image of the next file number on the PC card.

Camera Tab



The Camera tab is used to set the camera modes and the functions for recording.

AE Mode:
Selects the exposure mode of the camera (AUTO/MANUAL selectable).

FL Mode:
Selects the fluorescent mode of the camera (ON/OFF selectable).

Shutter:
Sets the electronic shutter speed in MANUAL exposure mode.

CAM (camera) Gain:
Sets the master gain of the camera in MANUAL exposure mode.

EV Comp. (compensation):
Sets the exposure compensation value in AE-AUTO or FL mode.

Sensitivity:
Sets the ISO sensitivity in AE-AUTO mode (ISO 100/200/400 selectable).

AE Lock:
Fixes the exposure time ON or OFF in AE-AUTO or FL mode.

AE Detect:
Changes the size of the AE window (SPOT/MID/LARGE selectable).

WB (white balance) Mode:
Selects the white balance mode of the camera (AUTO/MANUAL selectable).

RED Gain:
Adjusts the red gain manually.

BLUE Gain:
Adjusts the blue gain manually.

WHITE One Push button:
Activates the auto white balance mode (WB AUTO).

Sharpness:
Adjusts the sharpness of the image.

Locations and Functions of Controls (for Windows)

System Tab



The System tab is used to set the camera modes and the functions for recording.

Image Size:

Selects the size of the image to be recorded to the PC card (640 x 480/1280 x 960/2560 x 1920 selectable).

Image Quality:

Selects the recording format of the image – selectable from TIFF (uncompressed), JPEG High (JPEG low compression), JPEG Middle (JPEG middle compression), or JPEG Low (JPEG high compression).

E.Z. (easy) Focus:

Activates E.Z. Focus to facilitate focus operations in Camera LIVE mode.

Flash:

Selects the Flash mode when using a flash lamp or a slave unit – selectable from OFF, ON (EXT: OUTPUT), and ON (EXT: INPUT).

Gamma:

Selects the gamma compensation mode – selectable from ON (MS), ON (STD), and OFF.

Rotate:

Rotates the camera image by 180˚ – selectable from 0˚(normal image) or 180˚ (upside down).

Interval REC:

Selects the interval-recording mode to ON or OFF.

B&W (Black and White) mode:

Switches the Camera LIVE image from color to black and white. When ON is selected, both the Camera LIVE image and the captured image become monochrome.

OSD Color:

Selects the color of the characters used for menus, information, and messages displayed on the LCD (WHITE/BLUE/RED selectable).

ID Mode:

Selects whether or not to assign a user ID number to each image.

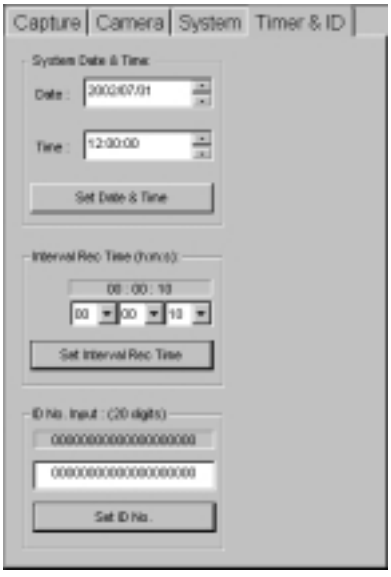
Histogram:

Displays a histogram on the LCD monitor of the CCU.

Display:

Displays information such as menus and focus indicator on the LCD monitor of the CCU.

Timer & ID Tab



The Timer & ID tab is used to set the clock and ID number for the camera.

System Date & Time:
Displays the date and time.

Set Date & Time button

After the date and time are set in the boxes, clicking this button transfers the values to the camera.

Interval Rec (recording) Time:
Sets the interval recording time for the camera..

Set Interval Rec Time button

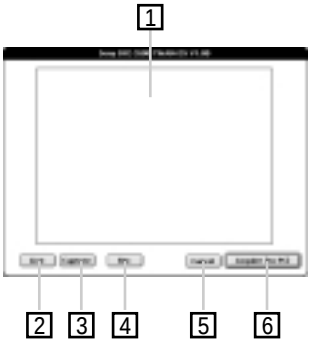
After the interval recording time is set in the box, clicking this button transfers the value to the camera.

ID No. Input:
Assigns a user ID number of up to 20 digits.

Set ID No. button

After the user ID number is set in the box, clicking this button transfers the ID number to the camera.

Locations and Functions of Controls (for Macintosh)



1 Image display area

Shows a live image, the index display, or a playback image.

2 LIVE button

Shows the live image on the LCD or video monitor.

3 Capture button

Captures a still image in the built-in frame memory of the camera.

4 Rec (record) button

Records a still image on the PC card inserted into the CCU.

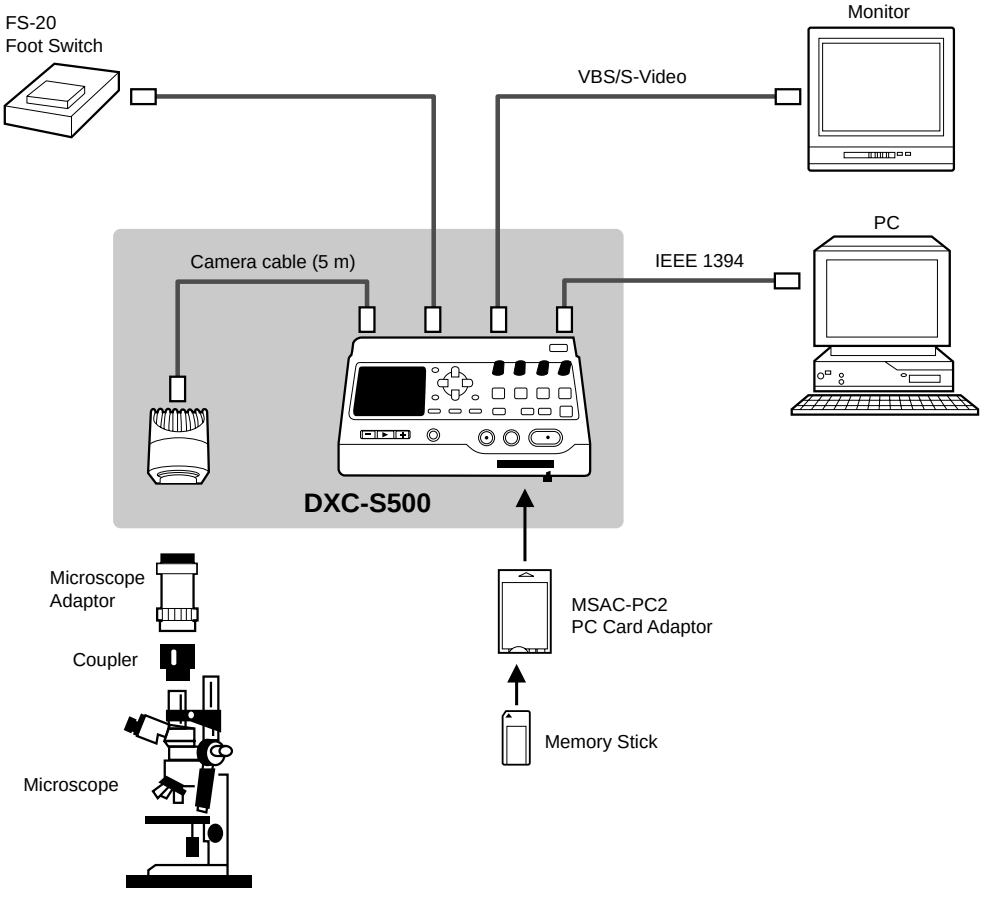
5 Cancel button

Clicking this button cancels the transmission of the image into the application software.

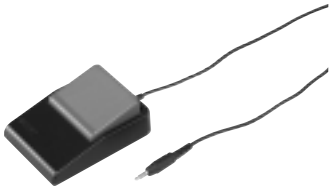
6 Acquire (to PC) button

Clicking this button transmits the images from the CCU into the application software. The transmitted image then appears in the application software window.

8. SYSTEM EXAMPLE



9. OPTIONAL ACCESSORIES



FS-20 Foot Switch



MSAC-PC2 Memory Stick Adaptor



MSA-8A/16A/32A/64A/128A Memory Stick



VMC-IL6615A/IL6635A IEEE 1394 cable (6-pin to 6-pin)

10. SPECIFICATIONS

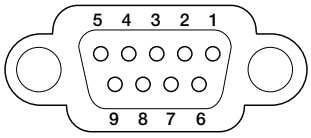
Imager	
Pick-up device	2/3 type IT (Interline Transfer) CCD, interlaced scan
Color filter	R, G, B primary color filter
Total picture elements (H x V)	5.24 mega-pixels (2658 x 1970)
Recording pixel elements (H x V)	2560 x 1920
Sensing area (H x V)	9.74 x 7.69 mm, square pixels
Camera	
Lens mount	C-mount
Flange back	17.526 mm in air
CCD cooling	Cooled to 5 °C (41 °F) at 0 °C to 35 °C (32 °F to 95 °F) ambient temperature
Bit depth	10-bit ADC
Sensitivity	ISO 100/200/400 selectable
Gain	Auto (AGC): 0 dB/6 dB/12 dB selectable Manual: 0 to 12 dB (1 steps)
Exposure control	AE: Auto/Manual selectable, AE lock function
AE area	Spot/Mid/Large
AE level	Variable
AE speed	Fast/Mid/Slow
Shutter speed	Auto/Manual (60 to 1/10000 s)/Electrical shutter
Gamma	ON (MS)/ON (STD)/OFF *1 switchable
White balance	Auto/Manual (R/B gain)
Detail level	Variable
EV compensation	-2 to 2 EV (0.25 EV steps)
Flash	Ext: Input mode/Output mode
User file	1/2/3 switchable (3 pattern memories)
Electrical zoom	2x (VGA size)
Fluorescent mode	ON/OFF
Black & White mode	ON/OFF
Interval recording	ON/OFF Max. 24 hours
User ID	ON/OFF Max. 20 digits
Picture rotate	0°/180° selectable
Live picture	
Image size (H x V)	640 x 480 pixels
Frame rate	30 fps/25 fps
Display	Digital interface and analog video output
LCD	
Size	3.5 type
Type	TFT color LCD made of low-temperature polysilicon
Resolution	200000 dots
Frame rate	NTSC: 30 fps, PAL: 25 fps
Image display	Live image or recording image (VGA size)

*1: MS: for Microscopy, STD: for Standard

Recording system	
Recording media	PC card-type removable HDD (PCMCIA-ATA Type II), Max. 5 GB Memory Stick (PC card adaptor required), Max. 128 MB CompactFlash (PC card adaptor required), Max. 330 MB SmartMedia (PC card adaptor required), Max. 128 MB
Max. recording pixel number (H x V)	2560 x 1920
Image data size (H x V)	3 modes: 2560 x 1920, 1280 x 960, 640 x 480
Recording format	TIFF (uncompressed), JPEG (compressed, three modes)
Software	
TWAIN driver	Bundled, for Windows/Macintosh
Supported OS	Windows 98SE/ME/2000/XP, Mac OS 8.6/9.0/9.1/9.2
Supported application software	Image-Pro Plus version 4.5 or later (for Windows) Photoshop version 5.0 or later (for Windows/Macintosh)
Control from PCs	Data transmission, live picture display, camera control
Output signals	
	IEEE 1394 (400 Mb/s), UYUV 16 bit/s VBS: 1.0 Vp-p, 75 Ω, sync negative Y: 1.0 Vp-p, 75 Ω C: NTSC 0.284 Vp-p, 75 Ω (without sync) PAL 0.3 Vp-p, 75 Ω (without sync)
General	
Operating temperature	-5 °C to 45 °C (23 °F to 113 °F)
Storage temperature	-20 °C to 60 °C (-4 °F to 140 °F)
Power requirements	100 to 240 V AC, 50/60 Hz
Power consumption	Max. 24 W
Dimensions (W x H x D)	CHU: 56 x 72 x 72 mm (2 1/4 x 2 7/8 x 2 7/8 inches) CCU: 270 x 163 x 79 mm (10 3/4 x 6 1/2 x 3 1/6 inches)
Mass	CHU: 300 g (11 oz) CCU: 1.8 kg (3 lb 15 oz)
Connectors	IEEE 1394 output (6-pin jack)*2 Video output (BNC) External trigger in (BNC) FS1 (stereo mini jack) AC in PC card (PCMCIE-ATA type II) S-Video output (Mini DIN 4-pin) Flash out (X) FS2 (D-sub 9-pin) Camera cable (10-pin)
Supplied accessories	
	TWAIN driver software (CD-ROM) Camera cable User's guide for TWAIN driver software Lens mount cap Operation manual

*2: IEEE 1394: Bus power not supported

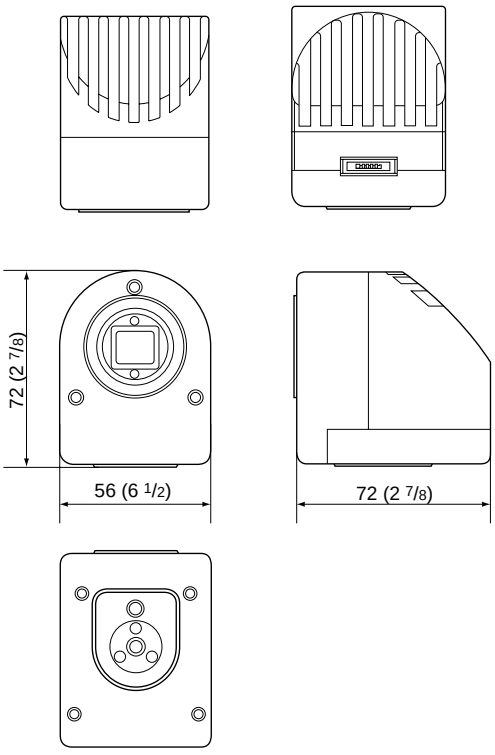
11. PIN ASSIGNMENT



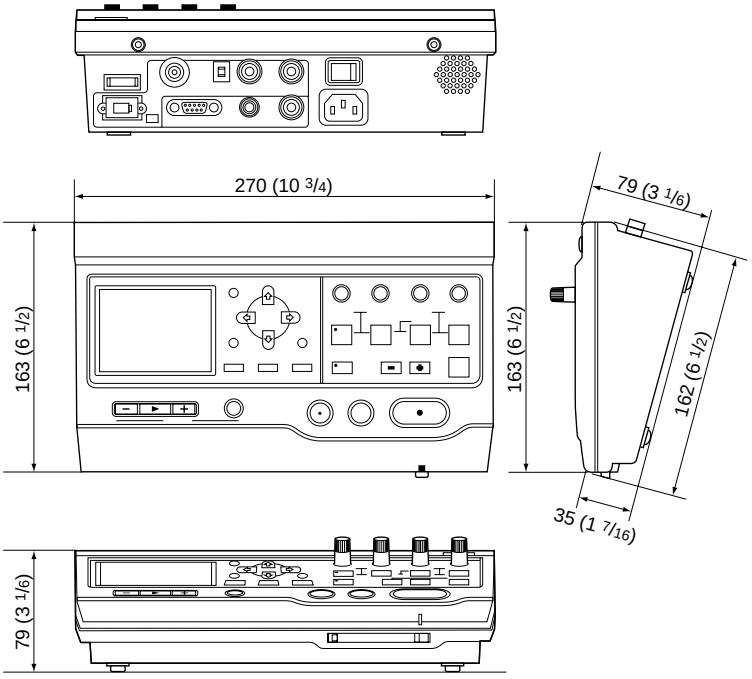
1	2	3	4	5	6	7	8	9
FOOT SW1	FOOT SW2	FOOT SW3	NC	NC	GND	NC	NC	NC

12. DIMENSIONS

Camera Head Unit (CHU)



Camera Control Unit (CCU)



Unit : mm (inches)

13. TECHNICAL APPENDIX

What's TWAIN?

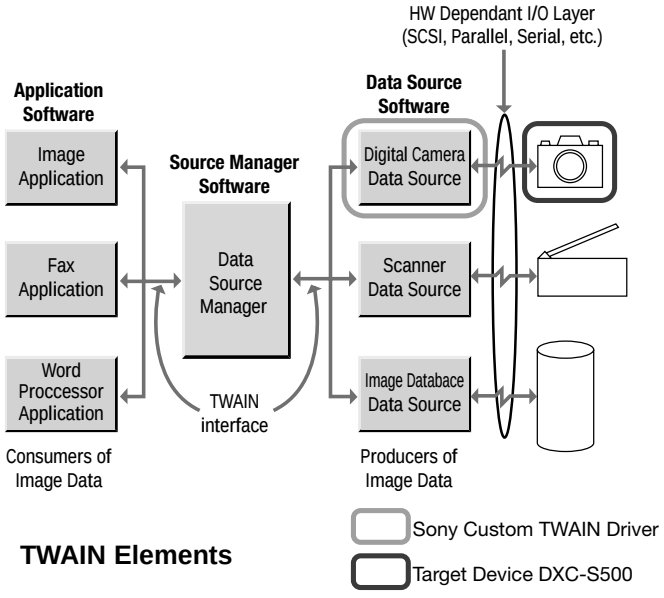
TWAIN is an open, industry-standard interface that directly acquires image data from devices such as scanners and digital cameras, without closing the application software.

The Elements of TWAIN

TWAIN defines a standard software protocol and API (application programming interface) for communication between the application software and the image-acquisition device (the source of the data).

The three key elements of TWAIN are:

- The application software – The application must be modified to use TWAIN.
- The Source Manager software – This software manages the communication between the application software and the source. The source code is provided in the TWAIN Developer's Toolkit.
- The Source software – This software controls the image-acquisition device and is programmed by the device developer to comply with TWAIN specifications.



TWAIN Elements

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